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Xu

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(54) **METHOD FOR SYNTHESIS OF ROXADUSTAT AND INTERMEDIATE COMPOUNDS THEREOF**

(71) Applicants: **Nanjing Cavendish Bio-Engineering Technology Co., Ltd.**, Nanjing (CN); **Yongxiang Xu**, Nanjing (CN)

(72) Inventor: **Yongxiang Xu**, Nanjing (CN)

(73) Assignees: **Nanjing Cavendish Bio-Engineering Tech. Co., LTD.**, Nanjing (CN); **Yongxiang Xu**, Nanjing (CN)

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C07D 217/26 (2006.01)

(52) **U.S. Cl.**
CPC ***C07D 217/26*** (2013.01)

(58) **Field of Classification Search**
CPC C07D 217/26
See application file for complete search history.

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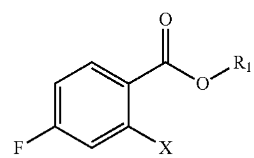
Primary Examiner — D Margaret M Seaman

(74) *Attorney, Agent, or Firm* — Thomas | Horstemeyer, LLP; Christopher Linder; Ragan Buckley

(57) **ABSTRACT**

The present application provides a method for synthesis of Roxadustat. A compound as represented by formula (VIII) is used as a raw material, and is reacted with phenol, a vinyl-containing ether, an acid, hydroxylamine, and then the product is reacted with glycine. In addition, the present application also provides intermediate compounds as represented by formula (IX), formula (XI), formula (XII), formula (IV), and formula (V) for synthesis of Roxadustat. Herein, details of the substituents involved in formula (VIII), formula (IX), formula (XI), and formula (XII) are stated in the description

(VIII)



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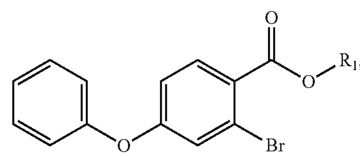

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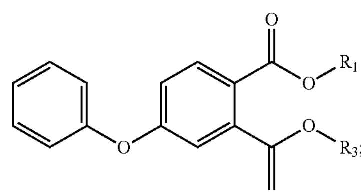
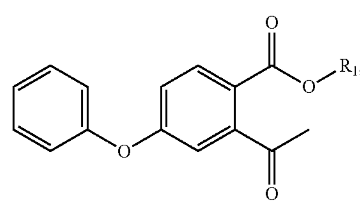


(XII)



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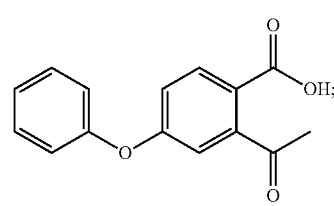

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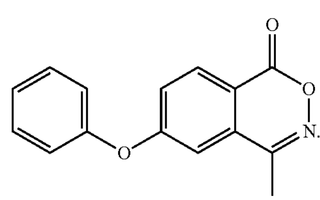


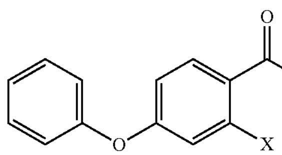
19 Claims, No Drawings

high-boiling organic solvent; herein, the high-boiling organic solvent is selected from toluene, xylene, and anisole;

the reaction is carried out in the presence of a base, herein, the base is selected from an organic base and an inorganic base, the organic base is one or more selected from triethylamine, N,N-diisopropylethylamine (DIEA), and n-butylamine; the inorganic base is one or more selected from potassium carbonate, sodium carbonate, potassium hydroxide and sodium hydroxide.

12. An intermediate compound for synthesis of Roxadustat, as shown in formula (IX), formula (XI), or formula (XII):

(IX)



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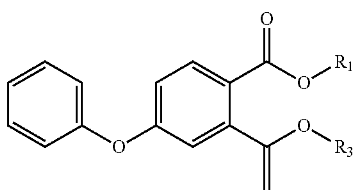
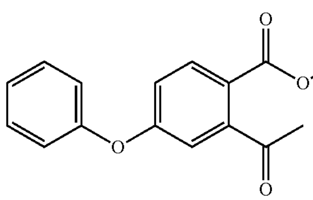


(XII)



wherein, R_1 in the formula (IX), formula (XI), and formula (XII) is each independently unsubstituted C_1-C_4 alkyl, C_1-C_4 alkyl substituted with C_1-C_4 alkoxy, C_1-C_4 alkyl substituted with unsubstituted aromatic group, C_1-C_4 alkyl substituted with unsubstituted aromatic oxy group, C_1-C_4 alkyl substituted with substituted aromatic group;

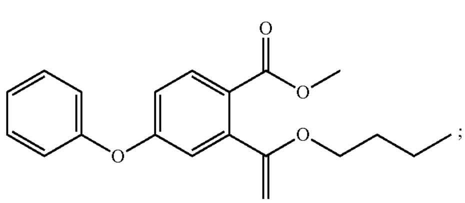
R_3 in the formula (XI) is each independently unsubstituted C_1-C_4 alkyl, C_1-C_4 alkyl substituted with C_1-C_4 alkoxy, C_1-C_4 alkyl substituted with unsubstituted aromatic group, C_1-C_4 alkyl substituted with unsubstituted aromatic oxy group, C_1-C_4 alkyl substituted with substituted aromatic group;

X in the formula (IX) is bromine, iodine or chlorine;

with the proviso that when X is bromine or chlorine, R_1 is not methyl.

13. An intermediate compound for synthesis of Roxadustat, selected from the following compound of formula (II), compound of formula (III), compound of formula (IV) or a salt thereof, and compounds of formula (V):

(II)



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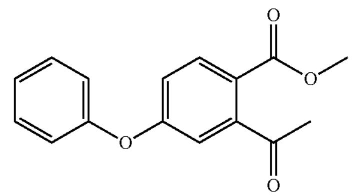

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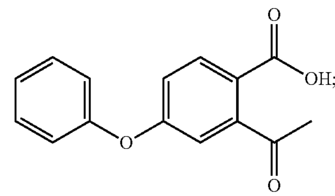

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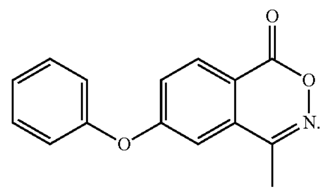

14. The method for synthesis of Roxadustat according to claim **1**, wherein the reaction in step (a) is carried out in the presence of a base; herein, the base is selected from alkali metal salts or alkali metal hydroxides, the alkali metal salts may be selected from alkali metal carbonates sodium carbonate, or potassium carbonate, the metal hydroxides may be selected from sodium hydroxide or potassium hydroxide.

15. The method for synthesis of Roxadustat according to claim **1**, wherein the reaction in step (a) is carried out in an aprotic organic solvent selected from dimethylformamide (DMF), and acetonitrile.

16. The method for synthesis of Roxadustat according to claim **1**, wherein the reaction in step (b) is carried out by using a metal catalyst and a phosphorus ligand; herein, the metal catalyst may be selected from $Pd(OAc)_2$, $Pd(PPh_3)_4$, palladium chloride, 1,1'-bis(diphenylphosphino)ferrocene] palladium dichloride, $NiCl_2(PPh_3)_2/Zn$, Cu/Al_2O_3, CoCl $(PPh_3)_3$; the phosphorus ligand can be selected from triphenylphosphine, tri-o-tolylphosphine.

17. The method for synthesis of Roxadustat according to claim **1**, wherein the acid hydrolysis reaction in step (c) is carried out in the presence of an acid; hererin, the acid is selected from an inorganic acid or an organic acid, the inorganic acid is hydrochloric acid, sulfuric acid, or phosphoric acid; the organic acid may be selected from acetic acid and trifluoroacetic acid.

18. The method for synthesis of Roxadustat according to claim **1**, wherein the reaction in step (d) is carried out in the presence of a base; herein, the base is selected from alkali metal salts, such as sodium carbonate, sodium acetate, or potassium carbonate.

19. The method for synthesis of Roxadustat according to claim **1**, wherein the reaction in step (e) is carried out in a high-boiling organic solvent; herein, the high-boiling organic solvent is selected from toluene, xylene, and anisole;

the reaction is carried out in the presence of a base, herein, the base is selected from an organic base and an inorganic base, the organic base is one or more selected from triethylamine, N,N-diisopropylethylamine (DIEA), and n-butylamine; the inorganic base is one or